

样 品 承 认 书

SAMPLE APPROVAL SHEET

安德普料号:
ADY PART NO.: _____

产品型号:
PRODUCT MODEL: ADY6812 _____

品名/规格:
NAME/SPEC: 80V100A _____

客户名称:
CUSTOMER NAME: _____

	Electrical	Material	Structure	Other
PASS				
Fail				
Other				
Director				

Remark:

ISSUE	CHECK	APPROVAL

QMS-21-001

V0

File Name:	ADY6812 80V100A Charger product Spec.				Confidential
File Attributes:	Product spec	Version:	V1.0	Pages:	2

ADY6812 High frequency smart charger specification



Model

80V100A

Version

V1.0

Compile:	Check:	Approve:
Qin Hai	Liao Zhaojun	Xu Liping

ADY-RD-002 V0

File Name:	ADY6812 80V100A Charger product Spec.				Confidential
File Attributes:	Product spec	Version:	V1.0	Pages:	3

Compile	Qin Hai	QE	Zhou Baoyuan
Hardware	Zhaojun Liao	Check	Zhaojun Liao
Software	Weijie He	Approve	Xu Liping

Change Record

Version No.	Update No.	Update date	Modify the description	Compile
V1.0		2021-10-05	First version	Qin Hai

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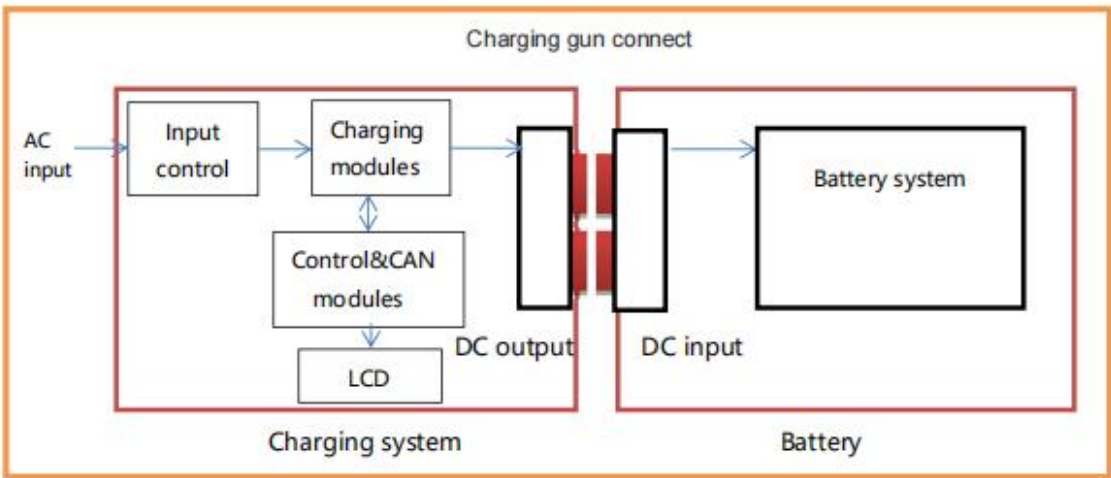
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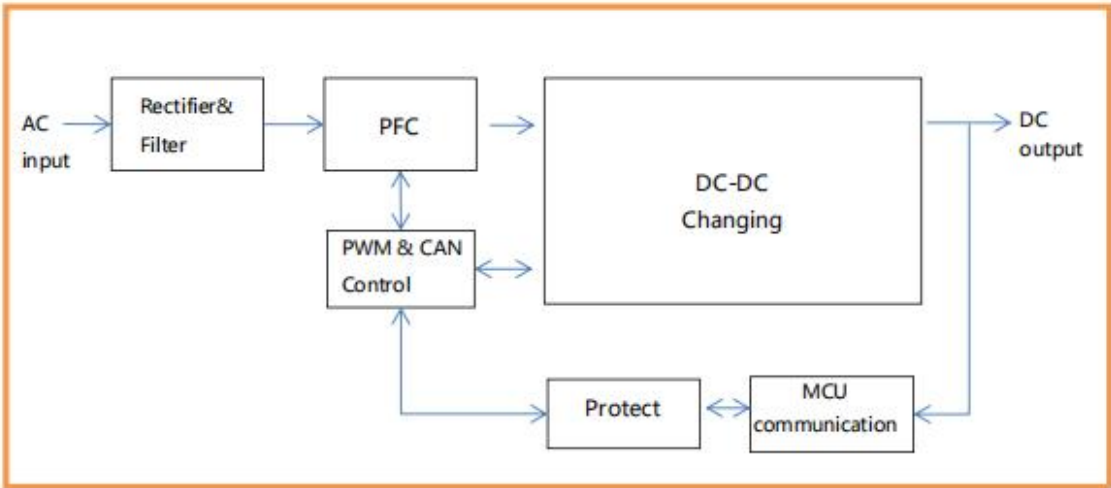
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1 Product Description

Welcome to use ADY6812 Charger made by ADY, this charger is an efficient, fast, and small-sized charging device. The charger is a three-phase AC 380V voltage input with a maximum output power of 11KW. It adopts forced air cooling and has the functions of input overvoltage, undervoltage, output overvoltage, undervoltage, overcurrent, output short circuit, and fan fault protection. At the same time, it meets RoHS and CE requirements.



System Architecture



Charging module topology framework

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2 Charger Electrical Parameters

2.1 Charger technical parameters

Item	Parameters	Remark
AC Input:		
Rated input voltage	380Vac	L1+L2+L3+PE
Input voltage range	324-520Vac (Full load) , 323-260Vac (Linearly reduced to half load)	/
Input connector	Industrial anti-riot plugs, crimping terminals	/
Input cable length	3000±100mm	/
Frequency range	47-63 Hz	/
Input current	≤16A Max	415Vac
Input power	12KW±10%	Rated input and output
Efficiency	≥93%	Rated input and output
Power factor	≥0.99	Rated input and output
DC Output:		
Rated output voltage	80V	/
Output voltage range	30-100V	Communication adjustment
Output current	0-130A	/
Output power	11KW	Max output power
Auxiliary supply output voltage	12V	BMS power
Auxiliary supply output current	5A	BMS power
Load regulation	±1%	/
Current precision	±1%	/
Voltage precision	≤±1%Vo (p-p)	20MHz bandwidth/peak to peak
MTBF	>80000H	25°C
Output cable、connector:		
Output connector	Cable with terminals	/
Output cable length	3000±100 mm	/
Output cable Spec	25 mm ²	/
Communication:		
communication method	CAN	Communicate with battery
Communication agreement	customer offer agreement	Communicate agreement with BMS battery
Charger:		
Size	L535*W220*H400 mm	Charger size (without foot pads)
Bare metal weight	31±2kg	For reference only, please refer to the actual measured value
Total Weight	34.3±2kg	For reference only, please refer to the actual measured value

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2.2 Safety Features

Item	Parameters	Remark
Isolation voltage	Input to the case 2500, 1min, leak current<30mA	Remove discharge tube test, require no arc, no breakdown
Isolation voltage	Input to output 2500; 1min, leak current<30mA	
Isolation voltage	Output to case 500; 1min, leak current<30mA	
Insulation resistance	Input to output $\geq 1\text{ M}\Omega$; output to case $> 5\text{ M}\Omega$	500Vdc test 25°C

2.3 Protect Function

Item	Unit	Parameters	Remark
Input overvoltage protection (software、 hardware)	Vac	≥ 520	Self-recovery, 325-520Vac (full load), 305-260Vac (linearly reduced to half load)
Input undervoltage protection (software、 hardware)	Vac	≤ 323	Self-recovery, 325-520Vac (full load), 323-260Vac (linearly reduced to half load)
Output overcurrent protection (software、 hardware)	A	$\geq 110\%I_o$	Self-recovery
Output undervoltage protection (software hardware)	Vdc	≤ 30	Self-recoverable, the output is guaranteed within the range of 30-50V, and the performance index is not guaranteed
Output overvoltage protection (software hardware)	Vdc	$\geq 110\%V_o$	Lock. Power off and restart.
Over-temperature protection (software)	°C	≥ 95	Outlet side radiator (test point)
Output short-circuit protection (software、 hardware)	/	Yes	Self-recovery
CAN communication interruption	/	Yes	Has Communication interruption and Intelligent shutdown after 5S' timeout

3 Working Environment

Item	Min	Typ	Max	Unit	Test condition
Working temperature	-20	25	40	°C	
Storage temperature	-40	25	70	°C	
Relative humidity	0	/	95	%	
Protection mode	/	IP21	/	/	

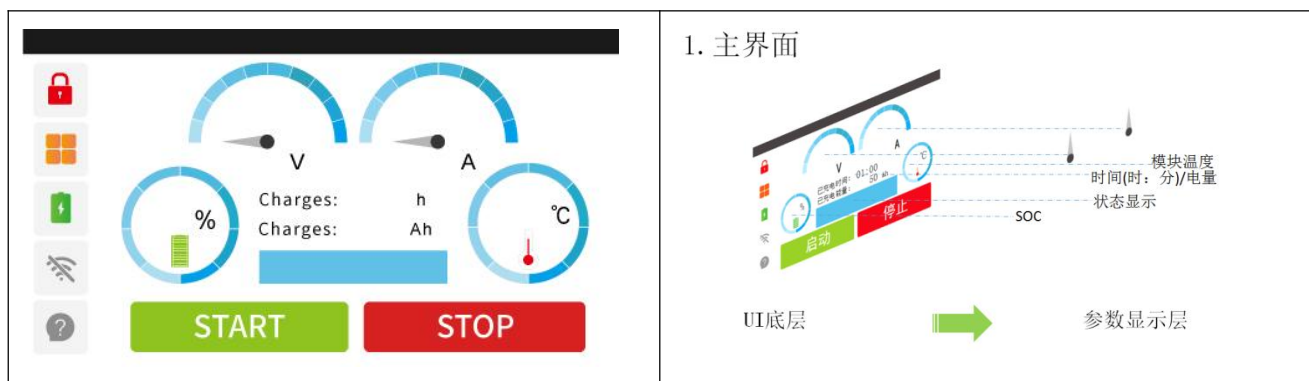
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Cooling method	/	Forced Air cooling	/	/	
Working altitude	/	2000m full load output	/	m	
Working noise	/	75	/	dB	1M from the charger

4 EMC & Safety

Item	Unit	Standard	Remark
CE	/	IEC61000_3_2 IEC61000_3_3 IEC61000_6_2 IEC61000_6_4	Refer to relevant certification standards

5 LCD Display panel and its contents



The above is the basic setting interface. For more detailed operation instructions, please refer to: Display UI Design.

Charger touch screen status display	State Description:
SelfChecking	Check the battery condition, whether charging is allowed, and whether the charger has faults itself
ChargerReady	External environment normal, It can be charged normally
Charging	Charging
Heating	State output in heating mode
StopCharging	The battery does not meet the condition of full battery

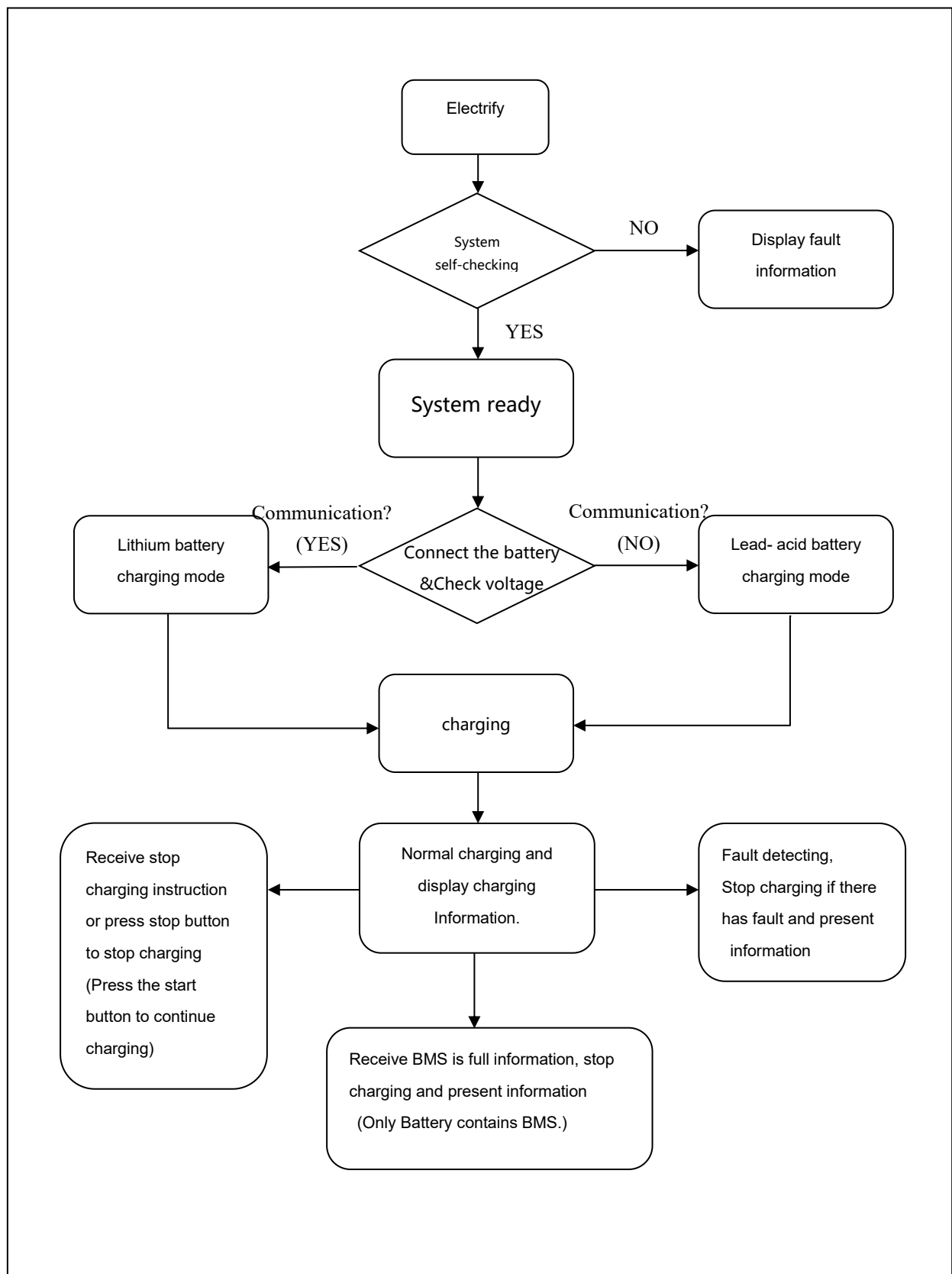
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Connect Bat	No battery voltage detected and Communication not established
No Bat Volt	Have communication connection, But the charger did not detect the voltage of the battery
Finished	SOC is equal to 100% or the charging voltage reaches 98% of the set voltage, and BMS sends the instruction to stop charging
Fan Fault	Module fan working abnormal
.AbnormalInput	The charger's input voltage is higher or lower than the preset value
Batt Req Err	The battery request voltage is greater than 10% allowed by the charger setting
Output OV	Output voltage is higher than the predetermined value
Output OC	Output current is higher than the predetermined value
Output OT	The charger modules temperature is higher than the setpoint
ModuleCommErr	Communication failure of charger modules
Battery OV	BMS detects battery abnormality and sends overvoltage command
Battery OC	BMS detects battery abnormality and sends overcurrent instruction
Battery OT	BMS detected battery abnormality and issued over temperature command
ChargeGun OT	The charger modules temperature is higher than the setpoint
BMS Comm Err	Communication between the BMS and the charger failed
ShortCircuit	Charger module output short circuit
Phase Err	Detected charger input phase loss/inverted fault
Battery UV	BMS detects battery abnormality and sends overvoltage command
Battery UT	BMS detected battery abnormality and issued over temperature command

Remark: 1.The above fault codes should do cooperation with BMS'send out,then can show.If the battery without BMS,some fault codes won't be displayed. Support for customer customization.

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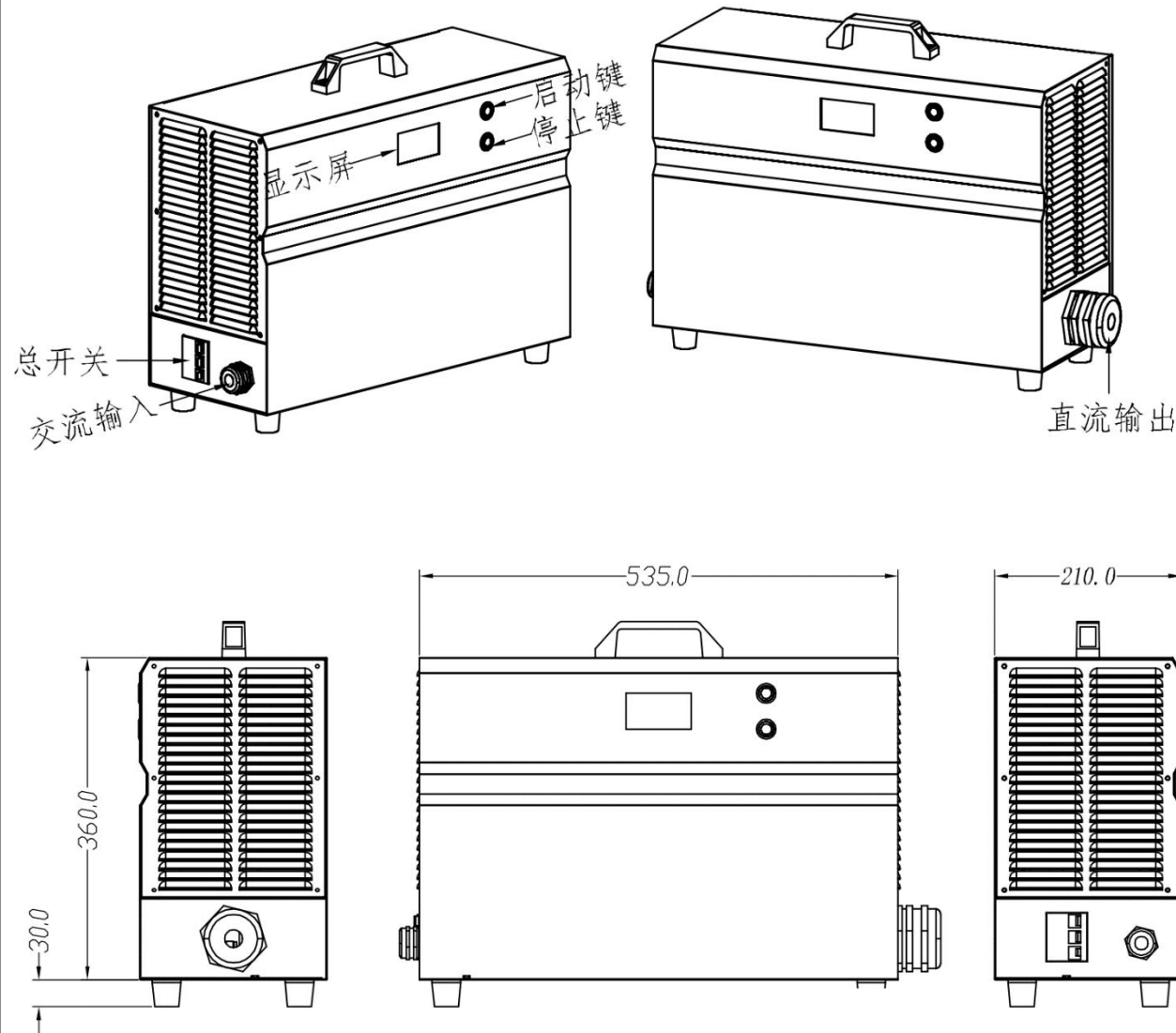
6 Charging Process



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7 Charger Structure (mm)

7.1 Structural dimensions



Overall size:

Length*width*height L535*W220*H400 mm (without foot pads)

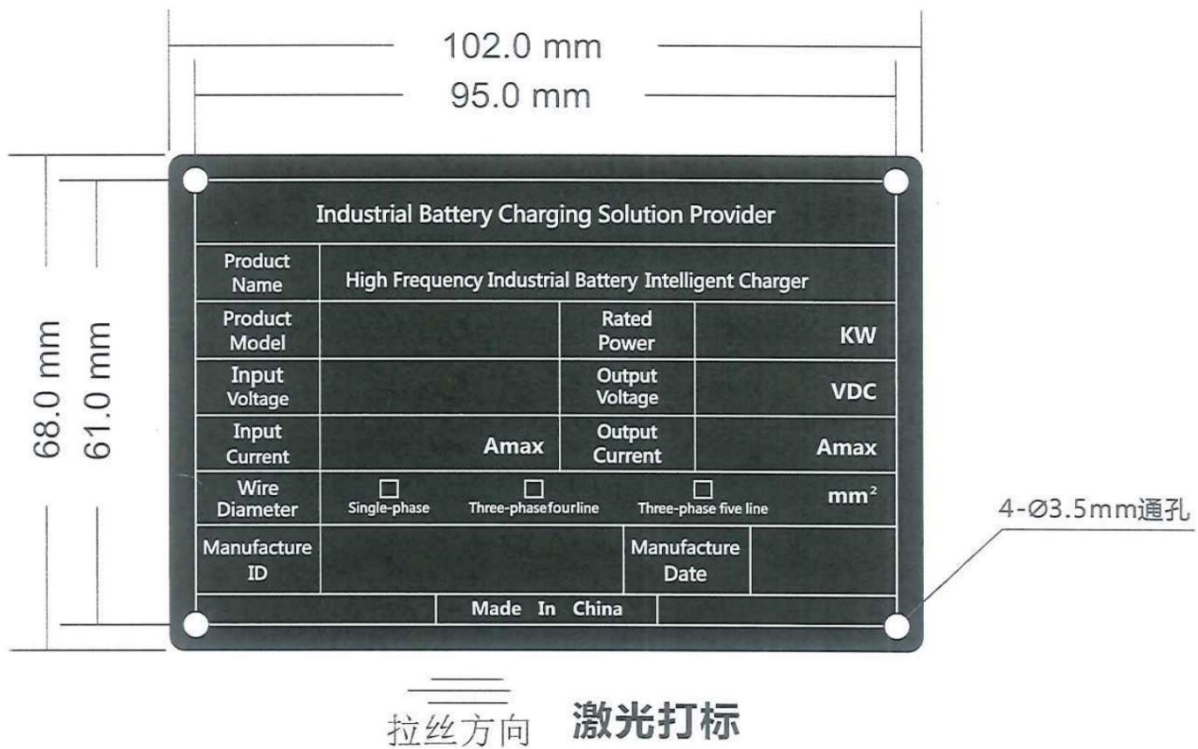
Length*width*height L535*W220*H430 mm (including foot pads)

Note:

1. The main color of the charger body is sand grain black RAL9005;
2. Paste our company's logo on the front of the body;
3. Other silk-screened white Chinese fonts;

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7.2 Schematic diagram of nameplate size

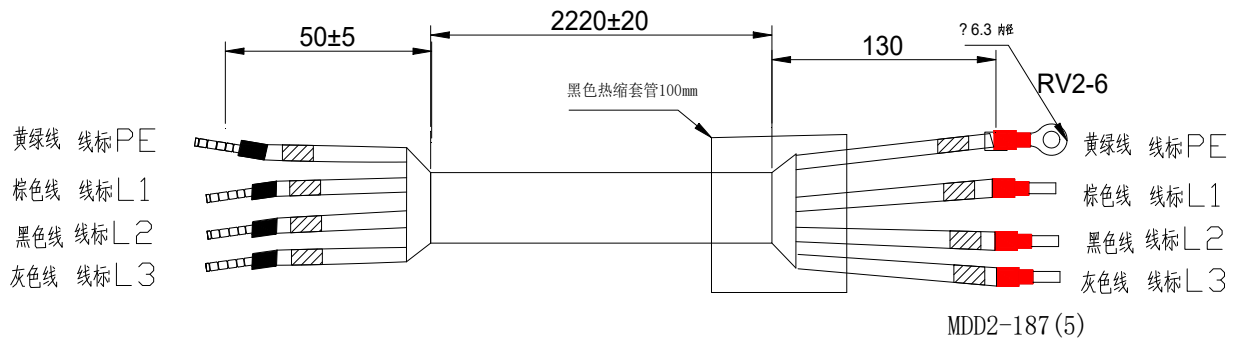


Note: For reference only, please refer to the actual product

7.3 Input cable and connector specifications

Phase	Specification	Remark
L1	2.5mm ²	Brown
L2	2.5mm ²	Black
L3	2.5mm ²	Grey
PE	2.5mm ²	Yellowish green

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Color: L1 (brown)、L2 (black)、L3 (grey)、PE (yellow green)

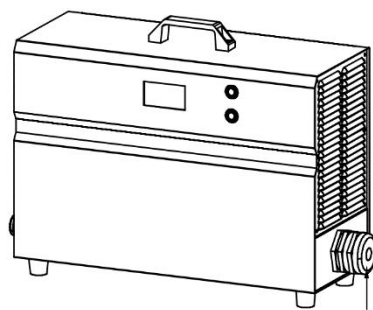
Note: This charger is a high-power leakage device. In order to ensure safety, please ensure that the PE terminal of the protective earth in the AC input is correctly connected to the earth before use;

7.4 Output cable and connector specifications

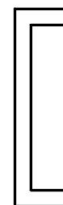
Adopt the crimping ring terminal with a wire diameter of 25 square millimeters.

8 Packaging

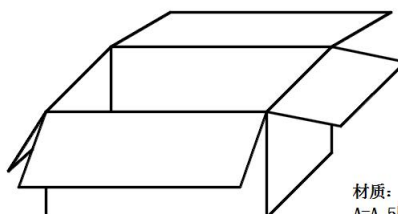
Carton (wooden box), pearl cotton, anti-static bag (for reference only, please refer to the actual packaging)



充电机



珍珠棉*4PCS



SHI
PHC
Add: Buil

材质:
A=A 5层瓦楞纸

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9 APPENDIX